Huawei Access Node YEL-ACC-67 QoS Congestion Event SOP (Major)

Node: YEL-ACC-67 | **Vendor:** Huawei | **Severity:** Major | **Alarm Code:** ALARM_4502

Description

The specified failure domain of QoS Congestion Event (Microbursts, interface tail drops, QoS queue starvation) on the Huawei Access Node YEL-ACC-67 at the Yelahanka Sub-station is precipitated by the inherent limitations of the Huawei's Carrier Ethernet (CE) architecture in managing high-traffic bursts and Quality of Service (QoS) priority settings. The Huawei's CE devices employ the IEEE 802.1Q-2014 standard for Ethernet VLANs and the IEEE 802.1p standard for traffic prioritization. However, the device's scheduling algorithm and buffer management mechanism may lead to microbursts and QoS queue starvation under high-traffic conditions. Furthermore, the device's interface tail drop mechanism may also contribute to congestion events. Hence, it is necessary to conduct a meticulous analysis of the QoS configuration, buffer settings, and scheduling algorithms to identify the root cause of the congestion event and implement corrective measures.

Pre-requisite Checks

- Ensure that the remote hands team is dispatched to the Yelahanka Sub-station for physical access to the Huawei Access Node YEL-ACC-67.

- Verify that the redundant link is functioning properly and there are no issues with the link's configuration or physical integrity.

- Check the power supply to the Huawei Access Node YEL-ACC-67 and ensure that it is functioning within the specified operating range.

Potential Root Causes

Cause	Probability
Misconfigured QoS priority settings	High
Buffer overflow due to high-traffic bursts	Medium
Scheduling algorithm misconfiguration	Low

Diagnostic Commands & Expected Outputs

```
Command: display current-configuration | include qos, Expected Output: display the current QoS configuration, Description: This command will display the current QoS configuration, including the QoS priority settings, which can help identify misconfigured priority settings.

Command: display buffer | include overflow, Expected Output: display buffer overflow information, Description: This command will display buffer overflow information, which can help identify buffer overflow due to high-traffic bursts.
```

```
Command: display scheduling | include algorithm, Expected Output: display scheduling algorithm information, Description: This command will display scheduling algorithm information, which can help identify misconfigured scheduling algorithms.
```

Resolution Steps

Step 1: Step: Verify and adjust QoS priority settings, Command: undo qos priority 1-7, Description: Verify the QoS priority settings using the display current-configuration | include qos command and adjust the settings as necessary to ensure proper traffic prioritization.

Step 2: Step: Adjust buffer settings, Command: buffer overflow-threshold 1000, Description: Adjust the buffer overflow threshold using the buffer overflow-threshold command to prevent buffer overflow due to high-traffic bursts.

Step 3: Step: Verify and adjust scheduling algorithm, Command: undo scheduling algorithm round-robin, Description: Verify the scheduling algorithm using the display scheduling | include algorithm command and adjust the algorithm as necessary to ensure proper traffic scheduling.

Post-Check Verification

```
Command: display current-configuration | include qos, Expected Output: display the current QoS configuration, Description: Verify that the QoS priority settings have been properly adjusted to ensure proper traffic prioritization.

Command: display buffer | include overflow, Expected Output: display buffer overflow information, Description: Verify that the buffer overflow threshold has been properly adjusted to prevent buffer overflow due to high-traffic bursts.

Command: display scheduling | include algorithm, Expected Output: display scheduling algorithm information, Description: Verify that the scheduling algorithm has been properly adjusted to ensure proper traffic scheduling.
```

Roll-back Procedures

- Step: Restore original QoS priority settings, Command: undo qos priority 1-7, Description: Restore the original QoS priority settings to their previous state using the undo qos priority 1-7 command if the fix fails.

- Step: Restore original buffer settings, Command: undo buffer overflow-threshold 1000, Description: Restore the original buffer overflow threshold to its previous state using the undo buffer overflow-threshold command if the fix fails.

- Step: Restore original scheduling algorithm, Command: undo scheduling algorithm round-robin, Description: Restore the original scheduling algorithm to its previous state using the undo scheduling algorithm command if the fix fails.